Exercise 38: Rescue Helicopter

Time Required:
Approx. 2 Hours

Interrelated Information:
Problem solving & Teambuilding

Objective:
Test a student's leadership ability in a team problem solving environment.

Instructor Directions:
Before class, cut out the strips of information below. Appoint a leader for the task. Give the leader the first strip of paper below. Distribute the other strips of paper among all the students. All the information plays into the solution, so all the strips of paper must be distributed. There are no limits on how the information can be shared. This GLP emphasizes finding the limiting factor in a schedule – a skill that officers will likely use again.

The team will have 1 hour to accomplish the task.
The second hour should consist of feedback from the team and from the instructor.

Points for observation:

- Did the leader effectively delegate tasks?
- Did the leader use all the tools at his disposal (paper, chalkboards)?
- Did the team stay focused and motivated to accomplish the mission?
- When was the mission articulated to the group & were they reminded of it when they got off track?
- What task oriented behaviors were exhibited by the leader and the team members?
    - Delegating
    - Summarizing
    - Directing
    - Defining
- What team-oriented behaviors were exhibited by the leader and the team members?
    - Encouraging/cheerleading
    - Drawing out quiet individuals
    - Applauding/rewarding participation
    - Keeping morale up
- Did the leader project confidence and poise?
- Did the leader adjust his leadership style based on the response from his subordinates and the problem solving environment (time, academic vs. physical, etc)

Information Slips

The commander has asked you to find out what is the earliest date and time that the rescue team can bring a downed pilot back to base. He expects the answer by 1400 hrs.

Tonight is a new moon.

The forecast is cloudy for the 11 th through the 15 th .

It is sunny today, there is not a cloud in the sky.

It will be a mostly clear night.

There is good daylight within the hangar with the doors open.

Avionics must be worked on inside the hangar.

It takes 30 min to move the helicopter either in or out of the hangar.

Today is August 10 th . The time is 1300 hrs.

If the weather is good, the maintainers can work by daylight until 2100 on the helicopter.

If the weather is clear and the moon is bright, the rescue team will not operate.

The extraction site is urban and considered too dangerous for a daytime mission.

Maintainers can only work 6 hrs/day on the helicopter in the hangar when it is cloudy or rainy because electricity needed for lighting and equipment is supplied intermittently on the base.

Electricity is usually available from 1000-1500 and from 2000-2300.

Sunrise will be at 0545 on August 11 th .

If the maintainers fix the broken avionics system in the rescue helicopter, the rescue team can execute the mission.

The rescue team needs 9 hours to plan and brief the mission.

The pilot and aircrew must complete 60 minutes of pre-flight checklists before taking off to start the mission.

If supply can procure replacement 50 Hz power cables, the maintainers can start troubleshooting the avionics system.

The time on the ground at the extraction site will be less than one hour.

The extraction site is located 50 min past Base Y.

It takes 2 hours to remove the avionics system from the helicopter for maintenance.

The maintenance group was notified about the broken helicopter at 1030 while the helicopter was on the ramp.

Supply drove to Base Y to get 50 Hz power cables at 1130.

Maintainers need 7 hours to troubleshoot and fix the avionics system.

Sunset will be at 2115 on August 11 th .

The diesel power generator can provide electricity to the avionics system while it is outside the helicopter for troubleshooting purposes, if 50 Hz cables are provided.

It takes 2.5 hours to reinstall the avionics system into the helicopter.

Base Y is 30 minutes away by helicopter.

Base Y is 50 km away by the local roads. The sergeant who left to procure the 50 Hz cables radioed to say that he was starting his return trip at 1230.

The diesel power generator consumes 200 liters/hour of fuel.

The diesel fuel tank is ¼ full. It has a capacity of 10,000 liters.

Maintainers can work without hangar lighting or daylight if they use flashlights and a diesel-fuelled light-all. All the tasks take twice as long. The light-all consumes 300 liters/hour of fuel.

Solution to Rescue Helicopter GLP

The rescue team can fly tonight to rescue the downed pilot. They will arrive back on station no later than 0540 on 11 Aug.

Rationale:

1030/10 Aug Maintenance notified about broken helicopter, begins moving helicopter into hangar

1100/10 Aug Helicopter in hangar, begin removing avionics system

1130/10 Aug Supply sergeant leaves to go to Base Y to pick up 50 Hz cables

1230/10 Aug Supply sergeant radios that he is starting his return trip from Base Y

1300/10 Aug Current date and time, when project officer is notified and tasked

1300/10 Aug Avionics system removed from helicopter, waiting on 50 Hz cables

1330/10 Aug (Assuming the sergeant makes it back in 1 hr from Base Y. This is how long the first leg took him) 50 Hz cables arrive

1330/10 Aug Maintenance fires up diesel power generator and begins working on avionics system

1400/10 Aug Briefing due to the general

1600/10 Aug Rescue crew comes off of crew rest, begins mission planning

2030/10 Aug Maintenance finishes troubleshooting and fixing the avionics system, begins reinstalling the system back into the helicopter.

2100/10 Aug No more daylight, electricity comes on in the hangar

2300/10 Aug Avionics reinstalled in the helicopter just before there is no more electricity, begins moving helicopter out onto the ramp using light-all

2400/10 Aug Fixed helicopter is on the ramp ready to fly

0100/11 Aug Rescue mission planned and briefed, begin pre-flight checklists

0200/11 Aug Rescue aircrew complete pre-flight checklists, take off

0230/11 Aug Rescue team arrives at Base Y

0320/11 Aug Rescue team arrives at the extraction site

0420/11 Aug Rescue team departs the extraction site

0510/11 Aug Rescue team arrives at Base Y

0540/11 Aug Rescue team arrives back to home base with downed pilot

0545/11 Aug Sunrise

Fuel calculations:

- Diesel 50 Hz power generator: 10 hours at 200 litres/hr = 2000 liters
- Diesel Light-All: 1 hour at 300 litres/hr = 300 liters
- Total diesel used = 2300 liters
- Capacity of tank is ¼ x 10,000 liters = 2500 liters
- 2300 < 2500, so no fuel limitation

Weather considerations:

- The rescue team can operate because there will be a new moon (no moon out), even though it is clear.
- Although electricity is interrupted to the base, the maintainers can adapt and overcome using the generator and light-all.